
Artificial Intelligence

Tut/Lab week 4

COSC2129

Outline

- Highlights:
 - Knowledge representation
 - Basic logic
 - Resolution
 - Inference
- Tutorial 4: KR&R I - Propositional Logic
- Q&A

Material

- Chapter 7, Book: Artificial Intelligence: A Modern Approach Global edition
- Book - TOC of Artificial Intelligence: A Modern Approach, 4th Global ed.
- AIMA: Pseudo code
- Good practices Python

Knowledge & Reasoning

- Knowledge representation: How do humans represent and store knowledge?
- Reasoning: How do humans do reasoning?
- How to transfer human knowledge to an agent?
- To be intelligent, an agent needs a **knowledge base** to store knowledge and an **inference engine** to do the reasoning.

Discussion

- Where to get knowledge to build a knowledge base?
- [MYCIN book](#)

Knowledge base

154 Knowledge Engineering

**NR

The new rule will be called RULE200.

If: 1-** THE ORGANISM IS A GRAM NEGATIVE ROD
and 2-** IT IS ANAEROBIC
and 3-** IT WAS ISOLATED FROM THE BLOOD
and 4-** YOU THINK THE PORTAL WAS THE GI TRACT
and 5-**

Then: 1-** IT IS PROBABLY A BACTEROIDES
On a scale of 1 to 10, how much certainty would
you affix to this conclusion?
** 9
and 2-**

This is my understanding of your rule:

RULE200

IF: 1) The site of the culture is blood, and
2) The stain of the organism is gramneg, and
3) The morphology of the organism is rod, and
4) The aerobicity of the organism is anaerobic, and
5) The portal of entry of the organism is GI
THEN: There is strongly suggestive evidence (.9) that the
organism is bacteroides

Okay? (YES or NO)

** YES

*[The knowledge engineer starts
the rule acquisition routine by
typing NR for New Rule.]*

*[user: carriage return with no
entry]*

*[user: carriage return with no
entry]*

*[Note that the original clause 1
has been expanded to separate
the two attributes, stain and
morphology.]*

FIGURE 7-2 Example of rule acquisition in the original (1974) MYCIN program. (User's input follows double asterisks.)

Knowledge Base

- Knowledge base (KB) in AI: captures human expert knowledge to support decision-making, problem-solving, and more
- KB: a **repository** of information.
- AI provides the mechanisms that enable machines/agents to “gain knowledge.” It allows them to acquire, process, and use knowledge to perform tasks that display “intelligent” behavior.

Logic

- Logic: a set of formal **languages** for representing **information** → draw conclusions
- Syntax: **sentences** in the language
- Semantics: “**meaning**” of sentences, i.e., **the truth** of a sentence.

Propositional Logic

- **Basic elements:**
 - Atomic sentences: $P, Q, R, \dots$ (uppercase letter)
 - T (always true), F (always false)
 - Connectives: $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$
- **Complex sentences** are constructed from simpler ones using connectives and parentheses.

$$P$$
$$P \wedge Q$$
$$P \vee Q$$
$$(P \vee Q) \Rightarrow R$$
$$((P \wedge Q) \Rightarrow (\neg R \vee (S \wedge P)))$$

Example

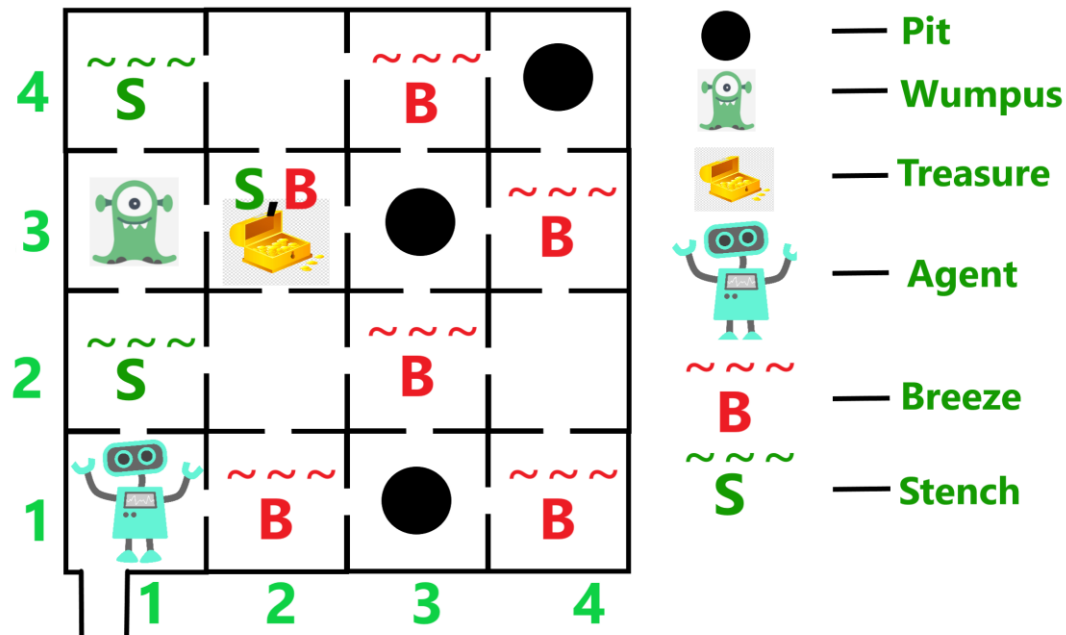
- P: "Grass is green"
- Q: "It is raining."
- R: " $2 + 5 = 5$ "
- $P \wedge Q$? $P \vee Q$?
- $P \rightarrow Q$
- ...
- p: "Close the door"
- q: "Is it hot outside?"
- Propositions
- True
- False
- Complex sentences
- Not propositions

Discussion

- Give an example of a KB with at least 5 sentences and 3 rules.
- Example: KB
 - Nam is smart
 - Nam studies hard
 - Nam passed the exam
 - Nam graduated
 - Nam gets a job after graduation
 - Nam is a successful person.
- Rules:
 - If One studied hard then One passed the exam.
 - If One passed the exam then One graduated.
 - If One gets a job then One is successful.

Example

- The Wumpus world



Truth Tables

P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	$(P \rightarrow Q) \wedge (Q \rightarrow R)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	F
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	T	T	T

□

- Truth Tables for Inference: enumerating all models (via truth table)
 - Complexity!

Entailment

- KB **entails** sentence α if and only if α is true in all worlds where KB is true:

$$KB \models \alpha$$

- Example:
 - The sentence (A) “The president was assassinated” entails (B) “The president is dead”.

Inference

- α is derived from KB by an inference algorithm i

$$KB \vdash_i \alpha$$

- Acquiring **new knowledge** from existing facts based on certain rules and constraints.

Inference

- To decide whether an entailment $\alpha \models \beta$ is true.
- 2 approaches:
 - Using Truth table
 - Applying inference rules (theorem proving approach)

Inference rule

$$\frac{\alpha_1, \alpha_2, \dots, \alpha_n}{\beta}$$

This general form of inference rules means “if we have proof that each α_i is true then this is extended to a proof that β is true.”.

A chained inference (i.e. possibly using more than one rule) from α to β (α infers β) is written:

$$\alpha \vdash \beta$$

Conjunctive normal form (CNF)

- It is a **conjunction** of one or more clauses, where each clause is **a disjunction** of literals

$(P1 \vee P2 \vee \dots \vee Pn) \wedge (Q1 \vee Q2 \vee \dots \vee Qm) \wedge \dots (\dots \vee \dots)$

Conjunctive normal form (CNF)

- Laws

$$\begin{cases} P \vee \bar{P} = T \\ P \wedge \bar{P} = F \end{cases}$$

$$\begin{cases} P \vee Q = Q \vee P \\ P \wedge Q = Q \wedge P \end{cases}$$

$$\begin{cases} (P \vee Q) \vee R = P \vee (Q \vee R) \\ (P \wedge Q) \wedge R = P \wedge (Q \wedge R) \end{cases}$$

$$\begin{cases} P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R) \\ P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R) \end{cases}$$

Conjunctive normal form (CNF)

- All complex sentences can be converted to CNF
 1. Replace $P \Leftrightarrow Q$ with $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$
 2. Replace $P \Rightarrow Q$ with $\neg P \vee Q$
 3. Push $\neg$ innermost
 - Replace $\neg\neg P$ with P
 - Replace $\neg(P \vee Q)$ with $\neg P \wedge \neg Q$
 - Replace $\neg(P \wedge Q)$ with $\neg P \vee \neg Q$
 4. Distribute $\vee$ over $\wedge$
 - Replace $P \vee (Q \wedge R)$ with $(P \vee Q) \wedge (P \vee R)$

Conjunctive normal form (CNF)

- Example: Convert to CNF

$$(P \Rightarrow Q) \vee \neg(R \vee \neg S)$$

$$(P \Rightarrow Q) \vee \neg(R \vee \neg S) \equiv (\neg P \vee Q) \vee (\neg R \wedge S)$$

$$\equiv ((\neg P \vee Q) \vee \neg R) \wedge ((\neg P \vee Q) \vee S)$$

$$\equiv (\neg P \vee Q \vee \neg R) \wedge (\neg P \vee Q \vee S).$$

Horn clause

- A CNF

$$\neg P_1 \vee \dots \vee \neg P_m \vee Q_1 \vee \dots \vee Q_n$$

Is equivalent to (Kowalski)

$$(P_1 \wedge \dots \wedge P_m) \rightarrow (Q_1 \wedge \dots \wedge Q_n)$$

- $n \leq 1$: Horn clause
- $n = 1$: $(P_1 \wedge \dots \wedge P_m) \rightarrow Q$

Resolution

- Resolution inference rule:

$$\frac{l_i \vee \dots \vee l_k, \quad m_1 \vee \dots \vee m_n}{l_i \vee \dots \vee l_{i-1} \vee l_{i+1} \vee \dots \vee l_k \vee m_1 \vee \dots \vee m_{j-1} \vee m_{j+1} \vee \dots \vee m_n}$$

where l_i and m_j are complementary literals.

e.g.,
$$\frac{P_{1,3} \vee P_{2,2}, \quad \neg P_{2,2}}{P_{1,3}}$$

- Resolution is sound and complete for propositional logic.

Resolution

- Resolution is a valid inference rule producing a new clause implied by two clauses containing **complementary literals**
 - A literal is an atomic symbol or its negation, i.e., P , $\sim P$
 - Used to build a sound and complete theorem prover!

Resolution inference rule

$$\frac{(\alpha \vee \neg\beta) \wedge (\gamma \vee \beta) \text{ premise}}{(\alpha \vee \gamma) \text{ conclusion}}$$

Proof by refutation

- A resolution refutation method is proof by contradiction using resolution
 - A resolution proof that derives the **empty clause**, as it shows that the input set of clauses F is **unsatisfiable**.
 - You begin by **assuming the opposite** of what you wish to prove, and then show that this “fact” would lead to a **contradiction**.

- Propositional Resolution example

1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

[illegible]

- Propositional Resolution example

1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

[illegible]

- Propositional Resolution example

1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

[illegible]

Practice

- Propositional Resolution example

Prove R

1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

Step	Formula	Derivation
1	$P \vee Q$	Given
2	$\neg P \vee R$	Given
3	$\neg Q \vee R$	Given
4	$\neg R$	Negated conclusion

Practice

- Propositional Resolution example

Prove R

1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

Step	Formula	Derivation
1	$P \vee Q$	Given
2	$\neg P \vee R$	Given
3	$\neg Q \vee R$	Given
4	$\neg R$	Negated conclusion
5	$Q \vee R$	1,2

Practice

- Propositional Resolution example

Prove R

1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

Step	Formula	Derivation
1	$P \vee Q$	Given
2	$\neg P \vee R$	Given
3	$\neg Q \vee R$	Given
4	$\neg R$	Negated conclusion
5	$Q \vee R$	1,2
6	$\neg P$	2,4

Practice

- Propositional Resolution example

Prove R

1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

Step	Formula	Derivation
1	$P \vee Q$	Given
2	$\neg P \vee R$	Given
3	$\neg Q \vee R$	Given
4	$\neg R$	Negated conclusion
5	$Q \vee R$	1,2
6	$\neg P$	2,4
7	$\neg Q$	3,4
8	R	5,7

Practice

- Propositional Resolution example

Prove R

1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

Step	Formula	Derivation
1	$P \vee Q$	Given
2	$\neg P \vee R$	Given
3	$\neg Q \vee R$	Given
4	$\neg R$	Negated conclusion
5	$Q \vee R$	1,2
6	$\neg P$	2,4
7	$\neg Q$	3,4
8	R	5,7
9	$\bullet$	4,8

Practice

- Propositional Resolution example

Prove R

1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

false $\vee$ R

$\neg R \vee$ false

~~false $\vee$ false~~

Step	Formula	Derivation
1	$P \vee Q$	Given
2	$\neg P \vee R$	Given
3	$\neg Q \vee R$	Given
4	$\neg R$	Negated conclusion
5	$Q \vee R$	1,2
6	$\neg P$	2,4
7	$\neg Q$	3,4
8	R	5,7
9	•	4,8

- Example: Prove P given the following

$$\neg A \vee \neg B \vee P \quad (1)$$

$$\neg C \vee \neg D \vee P \quad (2)$$

$$\neg E \vee C \quad (3)$$

$$A \quad (4)$$

$$E \quad (5)$$

$$D \quad (6)$$

Hand-on Practice

- [AI22 Tut4](#)

Part 2 discussion

- What if the order of checking sentences is different?

Part 3 discussion

Discussion

- Discuss about search and inference in the context of artificial intelligence.
 - Is inference a search?
 - How are they different?

Question?